

Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Blockchain Technology [CSE-4033]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Network A processes 10,000 transactions in 150 seconds, while Network B processes 8,000 transactions in 120 seconds. Calculate the throughput. Which network has a higher throughput, and what are their respective TPS? (5)
 - A)
 - B) Bitcoin allows users to transact directly without the need for intermediaries. Select the suitable type of blockchain and justify. (3)
 - C) The block header contains metadata about a block in the blockchain, including the previous block's hash, timestamp, nonce, and Merkle root. Identify how the nonce is used in the process of mining a block. (2)
- 2) A consortium of 15 banks deploys a permissioned blockchain using PBFT to manage interbank transactions. Initially, the network performs well with a few nodes, processing transactions efficiently. However, as more banks join the network, bottlenecks emerge. (5)
 - A) Examine and explain the bottlenecks in the PBFT protocol that affect its scalability.
 - B) Alice needs to sign an important contract online. She uses a digital signature based on RSA to sign the document and sends it to Bob. Apply asymmetric cryptography mechanism to ensure that Bob can verify the authenticity of the signature and confirm that the document has not been tampered with. What would happen if Alice's private key were compromised? (3)
 - C) As part of a blockchain startup, your task is to educate users and validators about the risks of the nothing-at-stake attack. Identify a communication plan to explain how this issue could affect blockchain forks and what actions validators can take to behave responsibly. (2)
- 3) Decentralized applications (dApps) are built on blockchain platforms like Ethereum and leverage the Ethereum Virtual Machine (EVM) to execute smart contracts. If a user creates a complex decentralized application (dApp), examine how the EVM handles multiple transactions triggered by the app. (5)
 - A)
 - B) When gas fees are high on the Ethereum network, it can make transactions expensive, especially for dApps requiring multiple interactions with smart contracts. If gas fees are high, identify steps taken by a dApp developer to reduce transaction costs. (3)

- C) The dApp allows voters to cast their votes for a list of candidates. Each voter is registered in advance, and votes are stored on the blockchain to ensure transparency and immutability. The goal is to optimize the contract to handle large numbers of votes without excessive gas fees. Identify the method to reduce gas consumption for the voting application. (2)
- 4) A company hires a freelance web developer to create a website. They sign a traditional contract that outlines the project scope, deadlines, payment terms, and dispute resolution mechanisms. The same company and freelancer decide to use a smart contract on a blockchain to automate the project agreement. Compare traditional contract enforcement methods with smart contract automation. (5)
- A) Artify, an online platform for buying and selling artwork, wants to implement an auction system that allows artists to sell their pieces directly to buyers through a transparent and automated process. Develop a smart contract for an online auction system. (3)
- C) InsureChain allows users to purchase insurance policies and file claims directly on the blockchain. Smart contracts automate the claims process by checking predefined conditions (such as external data feeds) to determine whether a claim should be paid. While the platform has the potential to revolutionize the insurance industry, it must address significant security threats to ensure user confidence and compliance. Analyze the threats in the Smart Contract for a Decentralized Insurance Platform. (2)
- 5) The choice of a hashing algorithm can have far-reaching consequences for data security and integrity. The shift from MD5 to SHA-256 in critical applications, especially in blockchain technology, illustrates the importance of selecting appropriate cryptographic tools to mitigate risks and enhance security. Compare and contrast different hashing algorithms (e.g., SHA-256 vs. MD5) in terms of security and efficiency. (5)
- A) HealthSecure opts for PoET, where each participant (healthcare providers, pharmacies, and patients) can validate transactions. Participants must use hardware with Trusted Execution Environments (TEEs) to participate in the consensus process. Analyze the trade-offs between PoA's efficiency and the decentralization provided by PoET. (3)
- C) In a Bitcoin network, a sudden increase could lead to blocks being mined much faster than the intended 10 minutes. As a result, after the 2016 blocks, the network would recognize the shorter block time and adjust the difficulty upward significantly. Identify the challenges faced by the PoW mechanism during a sudden increase in network hashing power. (2)

-----End-----